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```
mysql> INSERT INTO Products VALUES
-> (101, 'Laptop', 1),
-> (102, 'Rice Bag', 2),
-> (103, 'Notebook', 3),
-> (104, 'Mystery Item', 99); -- category_id 99 doesn't exist
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> SELECT *
-> FROM Products p
-> WHERE NOT EXISTS (
->   SELECT 1 FROM Categories c
->   WHERE p.category_id <> c.category_id
-> );
Empty set (0.00 sec)

mysql> SELECT *
-> FROM Products p
-> WHERE EXISTS (
->   SELECT 1 FROM Categories c
->   WHERE c.category_id = p.category_id
-> );
+-----+-----+-----+
| product_id | product_name | category_id |
+-----+-----+-----+
| 101 | Laptop | 1 |
| 102 | Rice Bag | 2 |
| 103 | Notebook | 3 |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

es incorrect

```
2 rows in set (0.00 sec)

mysql> SELECT DISTINCT s.name
-> FROM Students s
-> JOIN Enrollments e ON s.student_id = e.student_id
-> JOIN Courses c ON e.course_id = c.course_id
-> WHERE c.credits >= 4;
+-----+
| name |
+-----+
| Ravi |
| Meena |
+-----+
2 rows in set (0.00 sec)

mysql> CREATE INDEX idx_enroll_student ON Enrollments(student_id);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> CREATE INDEX idx_enroll_course ON Enrollments(course_id);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> CREATE INDEX idx_courses_credits ON Courses(credits);
Query OK, 0 rows affected (0.00 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> EXPLAIN SELECT name FROM Students WHERE student_id IN (
->   SELECT student_id FROM Enrollments WHERE course_id IN (
->     SELECT course_id FROM Courses WHERE credits >= 4));
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE | Students | NULL | ALL | PRIMARY | NULL | NULL | NULL | 2 | 100.00 | NULL |
| 1 | SIMPLE | Enrollments | NULL | ALL | idx_enroll_student,idx_enroll_course | NULL | NULL | NULL | 4 | 50.00 | Using where |
| 1 | SIMPLE | Courses | NULL | eq_ref | PRIMARY,idx_courses_credits | PRIMARY | 42 | employees.Enrollments.course_id | 1 | 66.67 | Using where; FirstMatch(Students) |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set, 1 warning (0.00 sec)

mysql>
mysql> EXPLAIN SELECT DISTINCT s.name FROM Students s
-> JOIN Enrollments e ON s.student_id = e.student_id
-> JOIN Courses c ON e.course_id = c.course_id
-> WHERE c.credits >= 4;
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE | c | NULL | range | PRIMARY,idx_courses_credits | idx_courses_credits | 5 | NULL | 2 | 100.00 | Using where; Using index; Using temporary |
| 1 | SIMPLE | e | NULL | ref | idx_enroll_student,idx_enroll_course | idx_enroll_course | 42 | employees.c.course_id | 1 | 100.00 | NULL |
| 1 | SIMPLE | s | NULL | ALL | PRIMARY | NULL | NULL | NULL | 2 | 50.00 | Using where; Using join buffer (hash join) |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set, 1 warning (0.00 sec)

mysql> |
```

Q2 The following sub-query returns incorrect results due to a missing

```
mysql> SELECT dept_id, COUNT(emp_id) AS emp_count
-> FROM Employees
-> GROUP BY emp_name;
ERROR 1055 (42000): Expression #1 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'Employees.dept_id' which is incompatible with sql_mode=only_full_group_by
mysql> SELECT dept_id, COUNT(emp_id) AS emp_count
-> FROM Employees
-> GROUP BY emp_name;
ERROR 1055 (42000): Expression #1 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'Employees.dept_id' which is incompatible with sql_mode=only_full_group_by
mysql> SELECT dept_id, COUNT(emp_id) AS emp_count
-> FROM Employees
-> GROUP BY dept_id;
```

dept_id	emp_count
1	2
2	1
3	1

```
3 rows in set (0.00 sec)

mysql> |
```

```
4 rows in set (0.01 sec)

mysql> UPDATE Employees
-> SET salary = salary * 1.10;
Query OK, 4 rows affected (0.01 sec)
Rows matched: 4 Changed: 4 Warnings: 0

mysql> SELECT * FROM Employees;
```

emp_id	emp_name	dept_id	salary
101	Arun	1	55000
102	Bala	2	49500
103	Chitra	3	66000
104	Divya	1	77000

```
4 rows in set (0.00 sec)

mysql> DELETE FROM Employees;
Query OK, 4 rows affected (0.01 sec)

mysql> INSERT INTO Employees SELECT * FROM Employees_backup;
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> UPDATE Employees
-> SET salary = salary * 1.10
-> WHERE dept_id = 1;
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2 Changed: 2 Warnings: 0

mysql> SELECT * FROM Employees;
```

emp_id	emp_name	dept_id	salary
101	Arun	1	55000
102	Bala	2	45000
103	Chitra	3	60000
104	Divya	1	77000

```
4 rows in set (0.00 sec)

mysql> SELECT * FROM Employees WHERE dept_id = 1;
```

emp_id	emp_name	dept_id	salary
101	Arun	1	55000
104	Divya	1	77000

```
2 rows in set (0.00 sec)
```

```

Chitra
Divya
+-----+
rows in set (0.01 sec)

mysql> ^C
mysql> ;
ERROR:
no query specified

mysql> SELECT e.emp_name
-> FROM Employees e
-> JOIN Departments d
-> ON e.dept_id = d.dept_id
-> WHERE d.location = 'Chennai';
+-----+
emp_name
+-----+
Arun
Chitra
Divya
+-----+
rows in set (0.00 sec)

mysql> EXPLAIN SELECT emp_name FROM Employees e
-> WHERE dept_id IN (SELECT dept_id FROM Departments WHERE location='Chennai');
+-----+
id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra
+-----+
1 | SIMPLE | Departments | NULL | ALL | PRIMARY | NULL | NULL | NULL | 3 | 33.33 | Using where
1 | SIMPLE | e | NULL | ALL | NULL | NULL | NULL | NULL | 4 | 25.00 | Using where; Using join buffer (hash join)
+-----+
rows in set, 1 warning (0.01 sec)

mysql>
mysql> EXPLAIN SELECT e.emp_name FROM Employees e
-> JOIN Departments d ON e.dept_id = d.dept_id
-> WHERE d.location = 'Chennai';
+-----+
id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra
+-----+
1 | SIMPLE | d | NULL | ALL | PRIMARY | NULL | NULL | NULL | 3 | 33.33 | Using where
1 | SIMPLE | e | NULL | ALL | NULL | NULL | NULL | NULL | 4 | 25.00 | Using where; Using join buffer (hash join)
+-----+
rows in set, 1 warning (0.00 sec)

mysql>

```

violation

```

-> student_id INT,
-> course_id VARCHAR(10)
-> );
Very OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Students VALUES
-> (1, 'Ravi'),
-> (2, 'Meena');
Very OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Enrollments (enroll_id, student_id, course_id)
-> VALUES (1, 105, 'CSA05');
Very OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Enrollments;
+-----+
enroll_id | student_id | course_id |
+-----+
1 | 105 | CSA05 |
+-----+
row in set (0.00 sec)

mysql> DROP TABLE Enrollments;
Very OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE Enrollments (
-> enroll_id INT PRIMARY KEY,
-> student_id INT NOT NULL,
-> course_id VARCHAR(10) NOT NULL,
-> CONSTRAINT fk_student FOREIGN KEY (student_id)
-> REFERENCES Students(student_id)
-> ON UPDATE CASCADE
-> ON DELETE RESTRICT
-> );
Very OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Enrollments (enroll_id, student_id, course_id)
-> VALUES (1, 105, 'CSA05');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('employees`.`enrollments`, CONSTRAINT fk_student FOREIGN KEY (student_id) REFERENCES students (student_id) ON UPDATE CASCADE ON DELETE RESTRICT ON UPDATE CASCADE)
mysql> INSERT INTO Enrollments (enroll_id, student_id, course_id)
-> VALUES (1, 1, 'CSA05');
Very OK, 1 row affected (0.01 sec)

mysql> select *from Enrollments
-> ;
+-----+
enroll_id | student_id | course_id |
+-----+
1 | 1 | CSA05 |
+-----+

```

```

mysql> CREATE VIEW DeptAvgSalary AS
-> SELECT dept_id, emp_name, AVG(salary)
-> FROM Employees
-> GROUP BY dept_id;
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE VIEW DeptAvgSalary AS
-> SELECT dept_id, AVG(salary) AS avg_salary
-> FROM Employees
-> GROUP BY dept_id;
ERROR 1050 (42S01): Table 'DeptAvgSalary' already exists
mysql> SELECT * FROM DeptAvgSalary;
ERROR 1055 (42000): Expression #2 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'employees.employees.emp_
is is incompatible with sql_mode=only_full_group_by
mysql> CREATE VIEW departmentAvgSalary AS
-> SELECT dept_id, AVG(Salary) AS avg_salary
-> FROM Employees
-> GROUP BY dept_id;
Query OK, 0 rows affected (0.01 sec)

mysql> select *from departmentAvgSalary ;
+-----+-----+
| dept_id | avg_salary |
+-----+-----+
| 1 | 60000.0000 |
| 2 | 45000.0000 |
| 3 | 60000.0000 |
+-----+-----+
3 rows in set (0.00 sec)

mysql> |

```

Q7. The GROUP BY query below produces wrong aggregation results. Debug the query and explain the error.

```

mysql> CREATE TABLE Departments2 (
-> dept_id INT PRIMARY KEY,
-> dept_name VARCHAR(50),
-> emp_id INT -- this is actually "head of department's employee id"
-> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> INSERT INTO Departments2 VALUES
-> (1, 'IT', 101),
-> (2, 'HR', 102),
-> (3, 'Finance', 103);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT *
-> FROM Employees e
-> JOIN Departments2 d
-> ON e.dept_id = d.dept_id AND e.emp_id = d.emp_id;
+-----+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary | dept_id | dept_name | emp_id |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Arun | 1 | 50000 | 1 | IT | 101 |
| 102 | Bala | 2 | 45000 | 2 | HR | 102 |
| 103 | Chitra | 3 | 60000 | 3 | Finance | 103 |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT e.emp_name, d.dept_name
-> FROM Employees e
-> JOIN Departments2 d
-> ON e.dept_id = d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Arun | IT |
| Bala | HR |
| Chitra | Finance |
| Divya | IT |
+-----+-----+
4 rows in set (0.00 sec)

mysql> |

```

Q9. Optimize the given multi-table SQL query by restructuring sub-

```
mysql> SELECT DISTINCT s.name
  -> FROM Students s
  -> JOIN Enrollments e ON s.student_id = e.student_id
  -> JOIN Courses c ON e.course_id = c.course_id
  -> WHERE c.credits >= 4;
+-----+
| name |
+-----+
| Ravi |
| Meena |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> CREATE INDEX idx_enroll_student ON Enrollments(student_id);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> CREATE INDEX idx_enroll_course ON Enrollments(course_id);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> CREATE INDEX idx_courses_credits ON Courses(credits);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> EXPLAIN SELECT name FROM Students WHERE student_id IN (
  -> SELECT student_id FROM Enrollments WHERE course_id IN (
  -> SELECT course_id FROM Courses WHERE credits >= 4));
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE | Students | NULL | ALL | PRIMARY | NULL | NULL | NULL | 2 | 100.00 | NULL |
| 1 | SIMPLE | Enrollments | NULL | ALL | idx_enroll_student,idx_enroll_course | NULL | NULL | NULL | 4 | 50.00 | Using where |
| 1 | SIMPLE | Courses | NULL | eq_ref | PRIMARY,idx_courses_credits | PRIMARY | 42 | employees.Enrollments.course_id | 1 | 66.67 | Using where; FirstMatch(Students) |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set, 1 warning (0.00 sec)
```

```
mysql> EXPLAIN SELECT DISTINCT s.name FROM Students s
  -> JOIN Enrollments e ON s.student_id = e.student_id
  -> JOIN Courses c ON e.course_id = c.course_id
  -> WHERE c.credits >= 4;
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE | c | NULL | range | PRIMARY,idx_courses_credits | idx_courses_credits | 5 | NULL | 2 | 100.00 | Using where; Using index; Using temporary |
| 1 | SIMPLE | e | NULL | ref | idx_enroll_student,idx_enroll_course | idx_enroll_course | 42 | employees.c.course_id | 1 | 100.00 | NULL |
| 1 | SIMPLE | s | NULL | ALL | PRIMARY | NULL | NULL | NULL | 2 | 50.00 | Using where; Using join buffer (hash join) |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set, 1 warning (0.00 sec)
```

```
mysql> |
```

Q10. Debug a given relational calculus expression that returns an empty

```
mysql> INSERT INTO Products VALUES
  -> (101, 'Laptop', 1),
  -> (102, 'Rice Bag', 2),
  -> (103, 'Notebook', 3),
  -> (104, 'Mystery Item', 99); -- category_id 99 doesn't exist
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT *
  -> FROM Products p
  -> WHERE NOT EXISTS (
  -> SELECT 1 FROM Categories c
  -> WHERE p.category_id <> c.category_id
  -> );
Empty set (0.00 sec)
```

```
mysql> SELECT *
  -> FROM Products p
  -> WHERE EXISTS (
  -> SELECT 1 FROM Categories c
  -> WHERE c.category_id = p.category_id
  -> );
```

```
+-----+-----+-----+
| product_id | product_name | category_id |
+-----+-----+-----+
| 101 | Laptop | 1 |
| 102 | Rice Bag | 2 |
| 103 | Notebook | 3 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> |
```